STPM Physics · Semester 3 · Chapter 24

# WAVE OPTICS

## 24.1 Huygens's Principle

### DEFINITIONS & LAWS

- **Wavefront** is a line or surface joining all points of a wave that are in phase (same displacement and same direction of motion).
- **Huygens's principle**: every point on a wavefront acts as a source of secondary spherical wavelets that spread out with the speed of the wave; the new wavefront after a short time t is the envelope (common tangent) of these wavelets.
- **Diffraction** is the spreading of a wave into the geometrical shadow when it passes the edge of an obstacle or through a gap; the points at the edge act as sources of secondary wavelets that spread into the shadow region.
- **Reflection and refraction**: construction with secondary wavelets gives the law of reflection (angle of incidence = angle of reflection) and Snell's law; on refraction the frequency is unchanged while the speed and wavelength change.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wave speed** | $v = f\lambda$ | $v$ = wave speed (m $s^{-1}$); $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Refractive index** | $n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$ | $n$ = refractive index of the medium (no unit); $c$ = speed in vacuum (m $s^{-1}$); $v$ = speed in the medium (m $s^{-1}$); $\lambda_0$ = wavelength in vacuum (air) (m); $\lambda$ = wavelength in the medium (m) |
| **Snell's law** | $n_1 \sin\theta_1 = n_2 \sin\theta_2$ | $n_1$, $n_2$ = refractive indices of the two media; $\theta_1$, $\theta_2$ = angles measured from the normal |
| **Wavelet radius** | $r = vt$ | Distance moved by a wavefront in time t, which is the radius of each secondary wavelet; $v$ = wave speed (m $s^{-1}$); $t$ = time (s) |

### TIPS TO ANSWER

- For a Huygens construction, draw several points on the wavefront, draw arcs of radius vt, then draw the common tangent as the new wavefront.
- State that the secondary wavelets have the same speed and frequency as the original wave.

- For diffraction, state that the edge points act as sources and the wavelets spread into the region behind the obstacle.
- For wavelength in a medium: state $\lambda = \frac{\lambda_0}{n}$, f unchanged, then substitute with units.

### SECTION A TRAPS

**Trap:** The frequency of light does not change when it enters glass or water. Options that say the frequency is multiplied or divided by n are wrong; only the wavelength and the speed change.

**Trap:** Plane wavefronts are not 'perfectly flat everywhere'. A small part of a spherical wavefront from a distant source is approximately plane; the radius of curvature is very large, not zero.

**Trap:** Wavefronts are drawn perpendicular to rays. Mixing up a ray (direction of travel) with a wavefront (surface of constant phase) loses the mark.

### TRICKS

**Trick:** Wavelength in a medium of refractive index n: divide by n (shorter), e.g. 600 nm in glass of n = 1.50 gives 400 nm. Never multiply.

**Trick:** If an option says the speed, frequency and wavelength all change on entering a new medium, eliminate it: f is fixed by the source.

## 24.2 Interference

### DEFINITIONS & LAWS

- **Principle of superposition**: when two or more waves meet at a point, the resultant displacement is the vector sum of the displacements of the individual waves at that point.
- **Interference** is the superposition of two or more coherent waves, producing a pattern of reinforcement (maxima) and cancellation (minima).
- **Coherent sources** are sources that emit waves of the same frequency (wavelength) with a constant phase difference.
- **Constructive interference** occurs when the waves meet in phase (path difference = nλ, n = 0, 1, 2, ...); **destructive interference** occurs when they meet in antiphase (path difference = (n + ½)λ).
- Two separate light bulbs do not give a stable interference pattern because the phase relationship of their light changes randomly and rapidly, so they are not coherent.
- On reflection from a medium of higher refractive index (denser medium) the wave undergoes a phase change of π (equivalent to a path change of $\frac{\lambda}{2}$); there is no phase change on reflection at a less dense medium.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Constructive interference** | $\Delta = n\lambda$ | $\Delta$ = path difference (m); $\lambda$ = wavelength (m); n = 0, 1, 2, ... |
| **Destructive interference** | $\Delta = (n + ½)\lambda$ | $\Delta$ = path difference (m); $\lambda$ = wavelength (m); n = 0, 1, 2, ... |
| **Phase difference** | $\varphi = \frac{2\pi}{\lambda} \times \Delta$ | $\varphi$ = phase difference (rad); $\Delta$ = path difference (m); $\lambda$ = wavelength (m) |
| **Intensity and amplitude** | $I \propto A^2$ | I = intensity (W $m^{-2}$); A = amplitude |
| **Maximum and minimum intensity** | $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$; $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$ | Two coherent waves of intensities $I_1$ and $I_2$ (W $m^{-2}$) |
| **Maximum and minimum amplitude** | $A_{max} = A_1 + A_2$; $A_{min} = \lvert A_1 - A_2 \rvert$ | Amplitudes add; $A_1$, $A_2$ = amplitudes of the two waves |

## TIPS TO ANSWER

- For path difference questions: divide $\Delta$ by $\lambda$. A whole number gives a bright (maximum); a half-integer gives a dark (minimum). State the conclusion with the value of $\frac{\Delta}{\lambda}$.
- For 'explain why no pattern' state: the sources are not coherent, i.e. the phase difference is not constant.
- For an intensity ratio: add and subtract the square roots of the intensities (amplitudes), then square.
- Mention any phase change of $\pi$ on reflection before deciding between maximum and minimum.

## SECTION A TRAPS

**Trap:** Half-integer path difference is destructive. A path difference of 2.5$\lambda$ is NOT 'close to a whole number, so constructive'; a path difference of 2.0$\lambda$ or 3.0$\lambda$ is bright.

**Trap:** Intensities do not simply add or subtract at the extremes. With $I_1 = 4I_0$ and $I_2 = I_0$: $\frac{I_{max}}{I_{min}} = \frac{(2 + 1)^2}{(2 - 1)^2} = 9$, not 3 (forgetting to square), not $\frac{5}{3}$ (which is $\frac{I_1 + I_2}{I_1 - I_2}$) and not 4.

**Trap:** Interference does not need waves of different types, and two lamps of the same colour are still incoherent. Coherence, not colour, is the requirement.

### TRICKS

**Trick:** Convert the path difference to a multiple of λ first $\frac{\Delta}{\lambda}$ and decide by inspection: integer = bright, half-integer = dark.

**Trick:** For $\frac{I_{max}}{I_{min}}$ with amplitude ratio r = $\frac{A_1}{A_2}$: ratio = $(\frac{r+1}{r-1})^2$. Intensity ratio 4 : 1 means r = 2, so the ratio is 9.

## 24.3 Two-slit Interference Pattern

### DEFINITIONS & LAWS

- **Young's double-slit experiment**: light from a single narrow slit (or a laser) illuminates two narrow slits $S_1$ and $S_2$ which act as coherent sources; the overlapping diffracted waves produce alternate bright and dark fringes on a screen.
- The bright fringe is formed where the path difference $S_2P - S_1P = n\lambda$; the dark fringe where $S_2P - S_1P = (n + ½)\lambda$.
- Fringes are equally spaced and of nearly equal brightness near the centre; the central fringe (n = 0) is bright because the path difference is zero.
- With white light, the central fringe is white and the other fringes are coloured (violet nearest the centre, red furthest) since fringe spacing ∝ λ.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Fringe spacing** | $x = \frac{\lambda D}{a}$ | x = fringe spacing (m); λ = wavelength (m); D = slit-to-screen distance (m); a = slit separation (m); valid for D ≫ a |
| **Path difference at a point** | $\Delta = \frac{ay}{D}$ | Δ = path difference (m); y = distance from the central fringe (m) |
| **Bright fringe position** | $y_n = \frac{n\lambda D}{a}$ | $y_n$ = position of the nth bright fringe (m); n = 0, 1, 2, … |
| **Dark fringe position** | $y_n = \frac{(n + ½)\lambda D}{a}$ | $y_n$ = position of the nth dark fringe (m) |
| **Angular position of bright fringe** | $a \sin\theta = n\lambda$ | θ = angular position of the nth bright fringe; a = slit separation (m); λ = wavelength (m) |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wavelength and spacing in a medium** | $\lambda' = \frac{\lambda}{n}$; $x' = \frac{x}{n}$ | In a medium of refractive index n; $\lambda'$ = wavelength in the medium (m); $x'$ = fringe spacing in the medium (m) |

## TIPS TO ANSWER

- Start from $x = \frac{\lambda D}{a}$, rearrange for the unknown, convert mm and nm to m, substitute with units, give 3 s.f.
- For 'distance between the mth and the nth fringes' use $(n - m)x$, not n x or m x separately.
- For effects of changing a quantity, state the proportionality: $x \propto \lambda$, $x \propto D$, $x \propto \frac{1}{a}$, then the new value.
- State assumptions: slits narrow, $a \ll D$ so that $\sin\theta \approx \tan\theta \approx \theta$.

## SECTION A TRAPS

**Trap:** Slit separation a is in the denominator: halving a doubles x. Options that give 'half' for halved a are wrong.

**Trap:** Unconverted units: 0.30 mm = $0.30 \times 10^{-3}$ m and 600 nm = $600 \times 10^{-9}$ m. A wrong power of ten is the most common slip.

**Trap:** Immersing the apparatus in a liquid DEcreases the fringe spacing ($x' = \frac{x}{n}$), it does not increase it. For n = 1.33, 1.60 mm becomes 1.20 mm, not 2.13 mm.

**Trap:** The angular position of the 3rd bright fringe uses m = 3, not m = 1; and with small angles the answer in degrees is very small (0.215°), not 3°.

## TRICKS

**Trick:** Ratio method: if only one quantity changes, $x_2 = x_1 \times$ (factor), no need to compute the full expression.

**Trick:** Check the answer is sensible: for visible light, a ~ 0.1–1 mm and D ~ 1–2 m, x is of the order of mm.

# 24.4 Interference in A Thin Film

## DEFINITIONS & LAWS

- **Thin-film interference** is the interference between the light reflected from the top surface and the light reflected from the bottom surface of the film; the two beams are coherent because they originate from the same incident wave.
- A phase change of π occurs on reflection when the light is reflected from a medium of higher refractive index; there is no phase change on reflection from a medium of lower refractive index.
- **Colours of a thin film in white light**: the film thickness varies, so different wavelengths satisfy the condition for constructive interference at different places, giving coloured bands.
- **Air wedge**: the air layer between two glass plates in contact at one edge; the fringes are straight, parallel and equally spaced, and the fringe at the line of contact is dark because of the π phase change at one reflection only.
- **Non-reflective coating**: a thin film of refractive index between those of air and glass; the reflected beams from the two surfaces interfere destructively so that reflection is reduced.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Thin film, one phase change: constructive** | $2nt = (m + ½)\lambda$ | Film of index n in air, one phase change at the top surface; t = film thickness (m); λ = wavelength in air (m); m = 0, 1, 2, ... |
| **Thin film, one phase change: destructive** | $2nt = m\lambda$ | Film of index n in air, one phase change at the top surface; m = 0, 1, 2, ... |
| **Thin film, no relative phase change: constructive** | $2nt = m\lambda$ | Both reflections with a π change, or neither (e.g. coating on glass with $n_{air} < n_{film} < n_{glass}$) |
| **Thin film, no relative phase change: destructive** | $2nt = (m + ½)\lambda$ | Both reflections with a π change, or neither (e.g. coating on glass with $n_{air} < n_{film} < n_{glass}$) |
| **Minimum coating thickness** | $t = \frac{\lambda}{4n}$ | Minimum coating thickness for destructive reflection of wavelength λ (m); n = refractive index of the coating |
| **Air wedge dark fringes** | $2t = m\lambda$ | Air wedge at normal incidence; dark fringes; t = air gap thickness (m); m = 0, 1, 2, ... |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Air wedge fringe spacing** | $x = \frac{\lambda}{2\theta}$ | Air wedge at normal incidence; x = fringe spacing (m); θ = wedge angle (rad) |

## TIPS TO ANSWER

- First decide the number of π phase changes at the two surfaces (list $n_1$, $n_2$, $n_3$ in order), then write the correct condition.
- State that the optical path difference is 2nt (the beam travels t down and t up inside the film) and that λ in the formula is the wavelength in air (the n is already included).
- Rearrange for t or λ, substitute, give the answer with unit, and name the order m used.
- For 'explain the colours', mention variation in thickness and the condition for each wavelength.

## SECTION A TRAPS

**Trap:** Check BOTH surfaces. For a soap film in air only the top surface gives the π change; for a coating (air → $MgF_2$ → glass) both surfaces do, so the conditions swap and $t = \frac{\lambda}{4n}$ gives destructive reflection.

**Trap:** Using $\frac{\lambda}{2n}$ instead of $\frac{\lambda}{4n}$: the minimum thickness for the non-reflective condition $2nt = \frac{\lambda}{2}$ is $t = \frac{\lambda}{4n}$, not $\frac{\lambda}{2n}$ or $\frac{\lambda}{n}$.

**Trap:** The wavelength in the film is $\frac{\lambda}{n}$. Do not use λ in the film for the path difference in addition to the factor n in 2nt (double counting).

## TRICKS

**Trick:** A film that is very thin ($t \to 0$) compared with the wavelength appears dark in reflection if there is one π change, so the first bright condition is at $t = \frac{\lambda}{4n}$.

**Trick:** Quick check on a soap-film answer: 560 nm light, n = 1.33 gives $t_{min} = \frac{560}{4 \times 1.33} = 105$ nm for strong reflection.

# 24.5 Diffraction by A Single Slit

## DEFINITIONS & LAWS

- **Diffraction** is the spreading of waves when they pass through a gap or round an obstacle; it is most marked when the width of the gap is comparable with the wavelength.
- A single-slit diffraction pattern has a wide, bright central maximum with weaker secondary maxima on both sides; the intensity of the secondary maxima falls rapidly.
- The pattern arises from interference between the secondary wavelets from different points across the width of the slit.
- The central maximum is twice as wide as each secondary maximum; the minima occur where the path difference between the two edges of the slit is a whole number of wavelengths.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Single-slit minima** | $a \sin \theta = m\lambda$ | m = 1, 2, 3, ... for the minima; a = slit width (m); θ = angle from the central axis (rad or °); λ = wavelength (m) |
| **Angular width of central maximum** | $2\theta_1$; $\sin \theta_1 = \frac{\lambda}{a}$ | $2\theta_1$ = angular width of the central maximum (rad); $\theta_1$ = angle of the first minimum; a = slit width (m); λ = wavelength (m) |
| **Width of central maximum on screen** | $w = \frac{2\lambda D}{a}$ | small angles; w = width of the central maximum (m); D = distance from slit to screen (m); λ = wavelength (m); a = slit width (m) |
| **Small-angle approximation** | $\theta \approx \frac{\lambda}{a}$ | for small angles; θ = angle of the first minimum (rad); λ = wavelength (m); a = slit width (m) |

## TIPS TO ANSWER

- Write the condition for minima first ($a \sin \theta = m\lambda$), then substitute with values converted to metres.
- For the width of the central maximum, double the distance from the centre to the first minimum: $w = \frac{2\lambda D}{a}$. State clearly 'width' versus 'half-width'.
- For 'what happens when a decreases', state: sin θ increases so the central maximum becomes wider (and less intense).
- A sketch of intensity against angle needs a labelled axis, a tall central peak that is twice the width of the others and small secondary peaks.

## SECTION A TRAPS

**Trap:** A narrower slit gives a WIDER pattern ($w \propto \frac{1}{a}$). Options saying the pattern becomes narrower or disappears are wrong.

**Trap:** Mixing up the angles: the first minimum is at $a \sin\theta = \lambda$, not at $a \sin\theta = \frac{\lambda}{2}$. The condition $a \sin\theta = m\lambda$ is for minima, not for maxima.

**Trap:** Forgetting the factor 2: the angular width of the central maximum is $2\theta_1$ (for example $2 \times 2.87° = 5.7°$), not $\theta_1$.

**Trap:** Do not confuse a (slit width) with d (slit separation or grating spacing) in the formula.

### TRICKS

**Trick:** If $a \gg \lambda$ the pattern is hardly spread; if $a < \lambda$ no minimum exists ($\sin\theta$ cannot exceed 1) and the light spreads over a wide region.

**Trick:** Width ratio: $\frac{w_2}{w_1} = \frac{\lambda_2}{\lambda_1}\frac{a_1}{a_2}\frac{D_2}{D_1}$ — use the proportionality instead of recalculating.

## 24.6 Diffraction Gratings

### DEFINITIONS & LAWS

- **Diffraction grating** is an optical device with a large number of equally spaced parallel slits (lines) of very narrow width; the diffracted beams interfere to produce sharp, bright maxima at definite angles.
- **Grating spacing d** is the distance between the centres of adjacent slits; $d = \frac{1}{N}$, where N is the number of lines per unit length.
- The maxima occur where the path difference between light from adjacent slits is a whole number of wavelengths.
- With white light, the zero-order maximum is white and each higher order is a spectrum with violet nearest and red furthest from the centre; the higher orders may overlap.
- Increasing the number of lines illuminated makes the maxima sharper and brighter; the angular positions of the maxima depend on d and $\lambda$ only.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Grating equation** | $d \sin \theta = m\lambda$ | m = 0, 1, 2, ... (order); d = grating spacing (m); θ = angle of the mth order from the normal (rad or °); λ = wavelength (m) |
| **Grating spacing** | $d = \frac{1}{N}$ | N = number of lines per metre ($m^{-1}$); d = grating spacing (m); for example 500 lines $mm^{-1}$ = 5.00 × $10^5$ lines $m^{-1}$, d = 2.00 × $10^{-6}$ m |
| **Highest order** | $m_{max} < \frac{d}{\lambda}$ | $m_{max}$ = largest integer less than $\frac{d}{\lambda}$ (because sin θ < 1); d = grating spacing (m); λ = wavelength (m) |
| **Total number of maxima** | $2m_{max} + 1$ | total number of maxima observed; $m_{max}$ = highest order |
| **Grating angle from screen** | $\tan \theta = \frac{y}{D}$ | y = distance on the screen from the central maximum (m); D = distance from grating to screen (m); then find sin θ from θ |

## TIPS TO ANSWER

- Calculate d from the number of lines per mm or cm FIRST, in metres, then apply $d \sin \theta = m\lambda$.
- For the maximum order, evaluate $\frac{d}{\lambda}$ and take the whole number part; state that sin θ cannot exceed 1.
- When the distance y on the screen is given, find $\theta = \tan^{-1}\frac{y}{D}$, then use sin θ in the grating equation (do not use $\frac{y}{D}$ directly for sin θ unless the angle is small).
- State the answer to 3 s.f. with the order m used.

## SECTION A TRAPS

**Trap:** Calculating d the wrong way round: for 600 lines per mm, $d = \frac{1}{6.00 \times 10^5} = 1.67 \times 10^{-6}$ m, not $600 \times 10^{-3}$ m. Remember d is the reciprocal of N.

**Trap:** Taking the highest order as $\frac{d}{\lambda}$ rounded up. For $\frac{d}{\lambda} = 3.40$ the highest order is 3, not 4; for 2.78 it is 2, not 3.

**Trap:** Using the 2nd order angle as twice the 1st order angle: it is the sine that is proportional to m, not the angle. For $d \sin \theta_1 = \lambda$ and $\sin \theta_2 = 2 \sin \theta_1$.

## TRICKS

**Trick:** Check sin θ ≤ 1 at once: if $\frac{m\lambda}{d} > 1$, that order does not exist.

**Trick:** More lines per mm means smaller d, therefore larger angles and fewer orders; use this to eliminate options quickly.

# 24.7 Polarisation

## DEFINITIONS & LAWS

- **Plane-polarised (linearly polarised) light** is light in which the electric field vibrations are in one plane only, perpendicular to the direction of propagation.
- Only transverse waves can be polarised; the polarisation of light shows that light is a transverse wave, whereas sound (a longitudinal wave) cannot be polarised.
- **Polariser** produces plane-polarised light from unpolarised light; **analyser** is a second polariser used to detect the plane of polarisation.
- **Malus's law**: the intensity of plane-polarised light transmitted through an analyser is proportional to the square of the cosine of the angle between the plane of polarisation of the light and the transmission axis of the analyser.
- **Brewster's law**: when light is incident at the polarising angle $\theta_B$, the reflected light is completely plane-polarised (vibrations perpendicular to the plane of incidence) and the reflected and refracted rays are perpendicular to each other.
- Unpolarised light passing through an ideal polariser is reduced to half of the incident intensity.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Malus's law** | $I = I_0 \cos^2 \theta$ | $I_0$ = intensity of plane-polarised light incident on the analyser (W $m^{-2}$); I = transmitted intensity (W $m^{-2}$); θ = angle between the plane of polarisation and the transmission axis |
| **Unpolarised light through a polariser** | $I_1 = \frac{I_{unpolarised}}{2}$ | $I_1$ = intensity after the first polariser (W $m^{-2}$); $I_{unpolarised}$ = incident intensity (W $m^{-2}$); then apply Malus's law for each further polariser |
| **Brewster's law** | $\tan \theta_B = \frac{n_2}{n_1}$; $\tan \theta_B = n$ | $\theta_B$ = Brewster (polarising) angle; $n_1$, $n_2$ = refractive indices of the first and second media; $\tan \theta_B = n$ for light incident from air |
| **Reflected and refracted rays at Brewster angle** | $\theta_B + \theta_r = 90°$ | $\theta_B$ = Brewster angle; $\theta_r$ = angle of refraction |

### TIPS TO ANSWER

- For unpolarised light: halve the intensity at the first polariser, then apply $I = I_0 \cos^2 \theta$ at each later polariser, using the angle between successive axes.
- For three polarisers, multiply the $\cos^2$ factors of successive pairs of axes (do not use the angle to the first axis for the third).
- For Brewster: state $\tan \theta_B = n$, substitute, and give $\theta_B$ in degrees; mention that the reflected ray is completely polarised.
- Name the type of wave: only transverse waves can be polarised.

### SECTION A TRAPS

**Trap:** Applying Malus's law to unpolarised light directly. The first polariser gives $\frac{I_0}{2}$ (average of $\cos^2$ is ½), and only then does $\cos^2 \theta$ apply: 100 W $m^{-2}$ → 50 W $m^{-2}$ → 37.5 W $m^{-2}$ at 30°.

**Trap:** Using the angle to the FIRST polariser for the third polariser. For crossed 1st and 3rd axes with a 45° polariser between them, each step is $\cos^2 45° = ½$, so the final intensity is $\frac{I_0}{2} \times ½ \times ½ = \frac{I_0}{8}$ (12.5%), not 0.

**Trap:** $\tan \theta_B = n$, not $\sin \theta_B = n$. Water n = 1.33 gives 53.1°; the complementary angle 36.9° is the refraction angle, not the Brewster angle.

**Trap:** Using $\cos \theta$ instead of $\cos^2 \theta$: at 60° the transmitted intensity is $\frac{I_0}{4}$, not $\frac{I_0}{2}$.

### TRICKS

**Trick:** Remember $\cos^2$ values: 30° gives $\frac{3}{4}$, 45° gives $\frac{1}{2}$, 60° gives $\frac{1}{4}$.

**Trick:** Inserting a polariser at 45° between crossed polarisers lets light through (a classic result); two crossed polarisers alone transmit zero.

## 24.8 Optical Waveguides

### DEFINITIONS & LAWS

- **Optical fibre** is a thin strand of glass or plastic consisting of a core surrounded by a cladding of lower refractive index; light is guided along the core by total internal reflection.

- **Total internal reflection** occurs when light travels from a denser to a less dense medium and the angle of incidence exceeds the critical angle.
- **Critical angle $\theta_c$** is the angle of incidence in the denser medium for which the angle of refraction in the less dense medium is 90°.
- Conditions for guiding: $n_{core} > n_{cladding}$ and the angle of incidence at the core–cladding boundary greater than $\theta_c$.
- **Attenuation** is the loss of power as the signal travels along the fibre, caused by absorption and scattering in the glass.
- **Modal dispersion** is the spreading (broadening) of a pulse because rays following different paths take different times; it is reduced in a very narrow core (single-mode) fibre or a graded-index fibre.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Critical angle** | $\sin\theta_c = \frac{n_2}{n_1}$ | $\theta_c$ = critical angle; $n_1$ = refractive index of the core; $n_2$ = refractive index of the cladding |
| **Numerical aperture** | $NA = \sin\theta_a = \sqrt{n_1^2 - n_2^2}$ | $\theta_a$ = maximum acceptance angle (from air); NA = numerical aperture (no unit); $n_1$ = refractive index of the core; $n_2$ = refractive index of the cladding |
| **Attenuation** | $\text{Attenuation} = \frac{10}{L}\log_{10}\frac{P_{in}}{P_{out}}$ | in dB per unit length; L = length of the fibre; $P_{in}$ = input power (W); $P_{out}$ = output power (W) |
| **Pulse transit time along axis** | $t = \frac{L}{v} = \frac{nL}{c}$ | t = time for a pulse along the axis (s); L = length of the fibre (m); $v = \frac{c}{n}$ = speed of light in the core (m $s^{-1}$); n = refractive index of the core; c = speed of light in vacuum (m $s^{-1}$) |

## TIPS TO ANSWER

- Identify $n_1$ (core, larger) and $n_2$ (cladding, smaller) before using $\sin\theta_c = \frac{n_2}{n_1}$; the ratio must be less than 1.
- For NA: substitute the two indices, give 3 s.f.; a numerical aperture is dimensionless.
- For an explanation of guiding, state both conditions: core index greater than cladding index AND angle of incidence at the boundary greater than the critical angle.
- For a time along the fibre, use $v = \frac{c}{n}$, not $v = c$.

## SECTION A TRAPS

**Trap:** Inverting the ratio: $\sin\theta_c = \frac{n_2}{n_1}$ (small over large). For core 1.50 and cladding 1.45, $\theta_c = 75.2°$, not 43.6°.

**Trap:** Saying the angle must be SMALLER than the critical angle for total internal reflection; it must be GREATER than $\theta_c$ (measured from the normal).

**Trap:** Forgetting that the core has the higher index; a core with lower index than the cladding cannot guide light.

**Trap:** Using the speed of light c for the transit time in the glass core: $t = \frac{nL}{c}$; for 2.0 km and n = 1.50, t = 10 µs, not 6.7 µs.

## TRICKS

**Trick:** Critical angle is large when the two indices are close; glass–air (n = 1.50) gives 41.8°.

**Trick:** NA = $\sqrt{(n_1 + n_2)(n_1 - n_2)}$ is quicker than squaring: $\sqrt{2.95 \times 0.05} = 0.384$.